

Three requirements on a relativistic home for MOND phenomenology

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Aether–scalar–tensor gravity (AeST) [Skordis & Złośnik, Phys. Rev. Lett. **127**, 161302 (2021)] is the only relativistic MOND-class theory known to reproduce the cosmic microwave background power spectrum, and it contains a single free function $\mathcal{F}(\mathcal{Y}, \mathcal{Q})$ that its authors leave unspecified. We show that specifying it is obstructed, and we extract from the obstruction three necessary conditions on *any* relativistic completion of MOND phenomenology. Each is established by an explicit construction that satisfies the previous conditions and fails the next. **(R1)** The free function must depend on the gradient of the *total* potential, not on the scalar’s own. With $\mathcal{F}(\mathcal{Y})$ the quasi-static reduction forces the anomalous acceleration $U(y)$ to be monotone increasing, hence to saturate at some $U \rightarrow s$, hence to produce a *constant* sunward anomaly $s a_0$ at every planetary distance. Perihelion precession then requires $s \leq 1.27 \times 10^{-5}$ while the radial acceleration relation requires $s \geq 0.435$ — incompatible by $1.2\text{--}3.4 \times 10^4$. There is no external-field relief: computed for this family rather than imported, it is $1.000000\times$. **(R2)** The cosmological background must not coincide with the deep-MOND end of the interpolation. Replacing $\mathcal{Y}$ by $Z \equiv J^\mu J_\mu$, the squared aether acceleration, satisfies R1 exactly — the reduction becomes AQUAL, the exponential interpolation becomes legal, and the 1 AU anomaly falls to $10^{-3458.7} \text{ m s}^{-2}$, voiding the gap — but $\bar{Z} = 0$ identically on Friedmann–Robertson–Walker (FRW) backgrounds, and any $\mathcal{F}$ performing the interpolation has $\mathcal{F}_Z(0) = K_B$ exactly, so the aether’s electric kinetic coefficient vanishes and the perturbation equation degenerates. **(R3)** Neither may be purchased with a split between the cosmological and local gravitational constants. Rescaling the free function repairs both R1 and R2 simultaneously, but forces $\bar{G}/G_N \leq 0.121$, which gives a primordial helium abundance $Y_p = 0.086$ against an observed 0.2449 ± 0.0040 (-42σ), and a gravitational radiation density *below that of the photons alone*, whose value is fixed by FIRAS with no gravity in the chain. Gravitational waves, by contrast, do not pay this price: the standard-siren amplitude’s sensitivity to the rescaling is exactly zero. We also record a structural identity: AeST’s action already contains an explicit $+(2 - K_B)a^\mu a_\mu$, so its apparent $c_4 = 0$ is an artifact of variables; and $c_T = 1$ holds exactly, for any free function.

I. SETUP

AeST’s action, verified against the authors’ source [1, 2], is

$$S = \int d^4x \frac{\sqrt{-g}}{16\pi\bar{G}} \left[R - 2\Lambda - \frac{K_B}{2} F^{\mu\nu} F_{\mu\nu} + 2(2 - K_B) J^\mu \nabla_\mu \varphi - (2 - K_B) \mathcal{Y} - \mathcal{F}(\mathcal{Y}, \mathcal{Q}) - \lambda(A^\mu A_\mu + 1) \right] + S_m[g], \quad (1)$$

with $\mathcal{Q} = A^\mu \nabla_\mu \varphi$, $\mathcal{Y} = (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi$, $F_{\mu\nu} = 2\nabla_{[\mu} A_{\nu]}$, and $J^\mu = A^\nu \nabla_\nu A^\mu$ the aether’s acceleration. Matter couples to $g_{\mu\nu}$ alone.

Throughout we use the acceleration scale

$$a_0 = \kappa c \sqrt{G\rho_\Lambda} = 9.3619 \times 10^{-11} \text{ m s}^{-2}, \quad (2)$$

with an alternative footing giving 1.1279×10^{-10} ; both are carried throughout. The coefficient $\kappa = \frac{1}{2}$ is *fitted, not derived*: three determinations give 0.551 ± 0.043 , 0.465 ± 0.076 and 0.537 ± 0.071 , combining to 0.529 ± 0.034 . Write $y = g_{\text{bar}}/a_0$ and $U(y) = u/a_0$ for the anomalous acceleration.

II. R1: THE FREE FUNCTION MUST EAT THE TOTAL GRADIENT

With $\mathcal{F}(\mathcal{Y})$, the quasi-static gradient sector diagonalises: $\Psi = \Psi_N + \chi$ with $\nabla^2 \Psi_N = 4\pi\hat{G}\rho$, and in spherical symmetry Gauss’s theorem gives the local algebraic law $u J_Y(u^2) = g_{\text{bar}}$ with $u = |\nabla\chi|$. The scalar’s own gradient *is* the anomalous acceleration. Reconstructing the free function from a chosen interpolation gives $J_Y(Y) = y(U)/U$, invertible iff U is strictly increasing; equivalently the longitudinal eigenvalue of $M_{ij} = J_Y \delta_{ij} + 2J_{YY} u_i u_j$ is positive under the same condition. A non-monotone U means both a multi-valued $\mathcal{F}$ and a longitudinal gradient ghost.

This is specific to AeST. In AQUAL [3] the free function depends on the total potential’s gradient and stability requires only $dg_{\text{obs}}/dg_{\text{bar}} > 0$ — satisfied by the exponential interpolation $\nu = 1/(1 - e^{-\sqrt{y}})$ [4], minimum 0.968. The same kernel is legal in one theory and fatal in the other, purely from which gradient the free function eats.

Monotonicity plus $U/y \rightarrow 0$ forces saturation, $U \rightarrow s$: a constant sunward anomaly $s a_0$ at every planetary distance. By the Gauss planetary equations a constant radial perturbation gives $\dot{\omega} = s a_0 \sqrt{1 - e^2}/(na)$; against anomalous-precession limits [5, 6] the worst planet re-

quires

$$s \leq 1.27 \times 10^{-5} \text{ (canonical)}, \quad 1.05 \times 10^{-5} \text{ (alt)}. \quad (3)$$

Applying $U(2) \geq 0.4$ to the family $J_Y = v/(1 - v/s)$, $v = \sqrt{Y}/a_0$, gives $s \geq 0.4348$; relaxing to a global fit (rms ≤ 0.15 dex on 3389 SPARC points [7], Υ in the Spitzer band) gives 0.157–0.294 depending on treatment. Hence

$$\boxed{\text{GAP: } 1.2\text{--}3.4 \times 10^4.} \quad (4)$$

Every candidate relief was computed and none helps. *The external-field effect contributes nothing:* derived for this family by two independent routes (an $\ell = 1$ penetration ODE and a flux bound requiring no perturbation theory), the relief on the saturated anomaly is $1.000000 \times$ — the external field screens *itself*, not the anomaly, and somewhere on the 1 AU sphere the anomaly is at least $s a_0(1 - 4 \times 10^{-9})$. A density-dependent a_0 *hurts*: with suppression f the constrained product is sf , and $f U_s(2/f) = 0.4$ gives 0.435 at $f = 1$ but 2.00 at $f = 0.1$, unsatisfiable below $f = 0.080$ because $U \leq \sqrt{y}$ caps the family. Solving the non-spherical problem for a Miyamoto–Nagai disc raises the required floor by 1.07–1.10. Refitting Υ buys $1.34 \times$.

III. R2: THE COSMOLOGICAL BACKGROUND MUST NOT BE THE DEEP-MOND POINT

R1 says: use the total gradient. AeST already contains the object that supplies it. With $D^\mu = q^{\mu\nu} \nabla_\nu \varphi$ one has the identity, verified three times independently and requiring no computer algebra,

$$\begin{aligned} 2(2 - K_B)J \cdot \nabla \varphi - (2 - K_B)\mathcal{Y} \\ \equiv +(2 - K_B)Z - (2 - K_B)|D - J|^2, \end{aligned} \quad (5)$$

with $Z \equiv J^\mu J_\mu$. AeST therefore already contains an explicit $+(2 - K_B)a^\mu a_\mu$: its apparent $c_4 = 0$ is an artifact of variables. Moreover $J^i = \nabla^i \Psi$ with coefficient exactly unity, with Z 's leading coefficient independent of φ , Φ and the aether perturbation — so Z is the total potential gradient squared.

Replacing $\mathcal{F}(\mathcal{Y}, \mathcal{Q})$ by $\mathcal{F}(Z, \mathcal{Q})$ then works exactly as R1 requires. The aether-longitudinal equation becomes $v = \nabla \Psi$ pointwise, the $(1 + J_Y)$ factor disappears, and the reduction is *AQUAL exactly*, with $\mu(g_{\text{obs}}) = J_Z(g_{\text{obs}}^2)$ and $\hat{G}$ unchanged. The exponential kernel becomes legal (min $dx/dy = 0.968$, against min $dU/dy = -0.032$ under $\mathcal{F}(\mathcal{Y})$ for the same kernel), and the 1 AU anomaly becomes $10^{-3458.7} \text{ m s}^{-2}$ — some 10^{3445} below the Sereno–Jetzer bound [8], replacing an $s a_0$ that sat $1279 \times$ above it. Further, $c_T^2 = 1$ exactly for any free function, and G_N is finite and equal to the unmodified value.

It fails on cosmology, for a reason that is the exact inverse of what protects $\mathcal{F}(\mathcal{Y})$. On FRW the aether is $A^\mu = (1, 0, 0, 0)$ and its acceleration vanishes, so $\tilde{Z} = 0$

identically — the cosmological background is the deep-MOND end of the interpolation. Expanding about it, the $O(\epsilon^2)$ action carries $[K_B - \mathcal{F}_Z(0)]Z^{(2)}$, so linear cosmology sees $K_B \rightarrow K_B - \mathcal{F}_Z(0)$. Any $\mathcal{F}$ that performs the MOND interpolation has $\mathcal{F}_Z(0) = K_B$ exactly, so the coefficient vanishes and the aether's electric evolution equation collapses to a constraint. $\mathcal{F}(\mathcal{Y})$ escapes only because $J_Y(\tilde{\mathcal{Y}} = 0) = 0$.

IV. R3: AND NEITHER MAY BE BOUGHT WITH A $\tilde{G}/G_N$ SPLIT

Both sector conditions constrain only the total Z -coefficient: static attraction needs it positive, vector no-ghost needs it below K_B . Rescaling the free function, $J \rightarrow sJ$, satisfies both at every acceleration for $s < K_B/(2 - K_B)$, and — the point — also repairs R2, since the rescaled normalisation gives $K_B - \mathcal{F}_Z(0) = K_B$ exactly on the deep-MOND branch. The window is $0 < s < 0.138$ at $K_B = 0.25$. The solar-system screening is untouched, being s -independent.

The price is that with $\mathcal{F}(Z)$ the non- $\mathcal{F}$ static Lagrangian is annihilated identically by $v = \nabla \Psi$, making the free function's normalisation the sole source of Newton's constant: $G_N = \hat{G}/s$. The no-ghost condition is therefore equivalent to $2\tilde{G} < K_B G_N$, i.e.

$$\tilde{G}/G_N \leq 0.121 \text{ } (K_B = 0.25), \quad 0.048 \text{ } (K_B = 0.10), \quad (6)$$

against AeST's 0.875 — an $8.3 \times$ to $20.7 \times$ split. It cannot be paid.

Big-bang nucleosynthesis. $Y_p = 0.086$ at the no-ghost ceiling against an observed 0.2449 ± 0.0040 : -42σ from the same calculation's $\tilde{G}/G_N = 1$ control. A closure-free bound assuming no nucleosynthesis model at all still gives $Y_p \leq 0.176$, i.e. -17σ .

The CMB, and this is an inconsistency rather than a tension. The gravitational sector's radiation density comes out at 0.2046 of the thermal photon density — below the photons alone, whose value FIRAS [9] fixes with no gravity anywhere in the chain. The equivalent $N_{\text{eff}} = -3.503$. The hard floor is missed by $4.9 \times$ ($K_B = 0.25$) to $12.2 \times$ ($K_B = 0.10$), and the ceiling sits 139σ from viability in cosmic-variance units.

Gravitational waves do not pay this price, and we state that against our own interest. The tensor sector is exactly Einstein–Hilbert: on FRW plus a transverse-traceless (TT) mode, $J^\mu = 0$, $Z = 0$, $F_{\mu\nu} = 0$ and $\mathcal{Y} = 0$ all hold identically and to all orders in h , so the dark sector carries no anisotropic stress for any free function, giving $c_T = 1$ and $\alpha_M = 0$ exactly. The standard-siren amplitude's sensitivity to s is then exactly zero: $\tilde{G}$ enters the strain and G_N the orbit, and they cancel in d_L^{GW} . GW170817 [10] places *no bound* on s . Two liabilities the same derivation creates are recorded but *not scored*: the inferred chirp mass rescales as $r^{3/5}$ ($4.13 M_\odot$ for GW170817 at the ghost edge, which needs the theory's own strong-field stellar structure to close), and the

tensor-channel orbital decay is $\dot{P}_b/\dot{P}_b^{\text{GR}} = r$, an $8\times$ deficit against B1913+16 [11]. We decline to score the second because the identical ledger applied to *unmodified* AeST would exclude it at 77σ — the tensor-only flux ledger is evidently incomplete, and the total radiated flux is not computed here. The usable result is a constraint on the shape of any repair: the missing channels must supply the same $1-r$ fraction at every orbital period, since $\dot{P}_b/\dot{P}_b^{\text{GR}}$ is measured to 0.16% in a 7.75 h orbit and to 1.3×10^{-4} in a 2.45 h one [12], and a dipole enters at relative order $(c/v)^2$ and cannot be period-independent.

V. TESTABLE PREDICTIONS

A no-go is tested by construction rather than by observation, but the derivations above carry four consequences that data can decide.

P1: $\alpha_M = 0$ and $c_T = 1$, exactly, for any free function. This is the strongest prediction here and the one least dependent on anything we have chosen. On FRW plus a TT mode, $J^\mu = 0$, $Z = 0$, $F_{\mu\nu} = 0$ and $\mathcal{Y} = 0$ hold identically and *to all orders in h* , so every non-Einstein invariant is h -independent and the dark sector’s TT stress is $T^x_x - T^y_y = 0$. The tensor sector is therefore exactly Einstein–Hilbert with $M_*^2 = 1/8\pi\tilde{G}$ constant. AeST and every construction in this paper are consequently indistinguishable from general relativity in gravitational-wave propagation — no running Planck mass, no modified friction, no c_T deviation, at any redshift and for any K_B , s or a_0 footing. GW170817’s $|c_T - 1| \lesssim 10^{-15}$ is passed trivially rather than fitted. A confirmed $\alpha_M \neq 0$ from LISA or Einstein Telescope standard sirens falsifies this entire class in one measurement, and cannot be absorbed by any choice of free function.

P2: the orbital-decay deficit must be period-independent. If the tensor-only flux ledger of Sec. IV is incomplete, as we argue, the missing channels must supply the same $1-r$ fraction of the flux at every orbital period. A dipole channel enters at relative order $(c/v)^2$ and therefore cannot be period-independent; the repair must renormalise the quadrupole coefficient itself. Any future binary showing a period-*dependent* deviation from the GR quadrupole formula excludes the repair, and with it the aether-vector rescaling.

P3: the $\mathcal{F}(\mathcal{Y})$ class predicts a constant sunward anomaly with a fixed planet ordering, $\tilde{\omega} = s a_0 \sqrt{1 - e^2}/(na)$, falling with semi-major axis in a way no fitted $\tilde{\omega}$ per planet can mimic, since a single s sets all of them. This is already excluded (Sec. II), which is *why* R1 exists; a future ephemeris solution reporting a coherent residual of this shape would revive rather than kill it.

P4: the $\mathcal{F}(Z)$ class predicts no solar-system deviation at all, at any measurable level: the 1 AU anomaly is $10^{-3458.7} \text{ ms}^{-2}$ and is s -independent. Any claimed solar-system MOND detection excludes it outright.

These are separate from framework-level predictions that this paper leaves untouched, since they belong to the normalisation (2) rather than to any completion: a hash-frozen wide-binary velocity ratio decided by Gaia DR4; a declining $a_0(z)$, which puts the MOND regime *off* at recombination at 0.0060 of its present value and is falsified by any robust a_0 evolution below $z \sim 5$; and $\nu_0 \leq 2.36 \times 10^{-6}$, i.e. no environmental variation of a_0 at a level current data could see.

VI. WHAT IS AND IS NOT CLAIMED

Claimed. The three requirements, each with an explicit construction that satisfies its predecessors and fails it. The identity (5) and its corollary that AeST’s $c_4 = 0$ is a choice of variables. That $c_T = 1$ holds exactly in AeST and in the $\mathcal{F}(Z)$ theory for any free function. That the monotonicity obstruction of R1 is genuinely escapable, and what escaping it costs.

Not claimed. That the requirements are sufficient — they are necessary conditions extracted from three failures, and a completion satisfying all three is not thereby viable. That $\kappa = \frac{1}{2}$ is derived; it is fitted, and must be quoted with its H_0 , the distance-free determination carrying an unpriced $\sim 7\%$ systematic. That dark matter is absent: Ω_{dm} is full here and the claim is “no dark-matter *particle*”. Whether the dark sector’s dust remains bound inside galaxies is unresolved and untouched by any of this — it is a $\mathcal{Q}$ -sector problem, and all three constructions modify the $\mathcal{Y}$ -sector.

Unaffected by any of the above, being independent of which completion carries them: the normalisation (2), the radial acceleration relation at 0.108 dex on 175 SPARC galaxies, weak lensing from 40 kpc to 2.2 Mpc with no dark component, the baryonic Tully–Fisher relation, and a hash-frozen wide-binary prediction decided by Gaia DR4.

What a viable completion must supply. A screening mechanism that is not a non-monotone interpolation (R1), whose free function’s argument does not vanish on FRW or whose derivative vanishes there (R2), and which does not separate the cosmological and local gravitational constants (R3). Superfluid constructions [13], in which the MOND force exists only inside a condensate phase and there is no interpolation function to have a deep-MOND end, satisfy R2 vacuously and are the obvious next candidate.

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Every number in this paper is produced by a committed self-checking script at <https://github.com/carlzimmerman/zimmerman-formula>; each prints numbered checks and exits non-zero on failure. Withdrawn claims are recorded, dated, in `RETRACTIONS.md` — including one in this paper’s own drafting, where a

gravitational-wave exclusion was written from an unverified number and corrected against the committed script before filing. The contribution here is the three require-

ments, the identity (5), and the constructions that establish them — not the underlying theory, which is Skordis and Złośnik’s.

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